

Lista 4 de G.A de Jeann Victor Batista.

a) $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \end{pmatrix} \quad \begin{matrix} X = 2 \\ Y = -1 \end{matrix}$

b)

$\begin{pmatrix} 1 & 0 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} X \\ Y \\ W \\ Z \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \\ 2 \\ 1 \end{pmatrix} \quad \begin{matrix} X = 4 \\ Y = 3 \\ W = 2 \\ Z = 1 \end{matrix}$

c)

$\begin{pmatrix} 1 & 0 & 0 & 0 & 6 \\ 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 1 & 1 & 2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} \cdot (X, Y, Z, W) = \begin{pmatrix} 6 \\ 3 \\ 2 \end{pmatrix} \rightarrow \begin{matrix} X = 6 \\ Y = 3 \\ Z + W = 2 \rightarrow Z = 2 - W \end{matrix}$

d)

$\begin{pmatrix} 1 & 0 & 3 & 1 \\ 0 & 1 & -1 & 2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & -1 \end{pmatrix} \cdot X, Y, Z = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad \begin{matrix} X - Z = 1 \\ Y - Z = 2 \end{matrix}$

e)

$\begin{pmatrix} 1 & 0 & 0 & -7 & 8 \\ 0 & 1 & 0 & 3 & 2 \\ 0 & 0 & 1 & 1 & -5 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 & -7 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 1 \end{pmatrix} \cdot a, b, c, d = \begin{pmatrix} 8 \\ 2 \\ -5 \end{pmatrix} \quad \begin{matrix} a - 7d = 8 \\ b + 3d = 2 \\ c + d = -5 \end{matrix}$

f)

$\begin{pmatrix} 1 & -6 & 0 & 0 & 3 & -2 \\ 0 & 0 & 1 & 0 & 4 & 7 \\ 0 & 0 & 0 & 1 & 5 & 8 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -6 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 & 4 \\ 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \cdot a, b, c, d, e = \begin{pmatrix} -2 \\ 7 \\ 8 \\ 0 \end{pmatrix}$

$a - 6b + 5e = -2$

$c + 4e = 7$

$d + 5e = 8$

$0 = 0$

$$2) a) \begin{cases} 3x - 2y = 1 \\ x + 3y = 9 \end{cases} \rightarrow \begin{Bmatrix} 3 & -2 & 1 \\ 1 & 3 & 9 \end{Bmatrix} \xrightarrow{l_2 \leftrightarrow l_1} \begin{Bmatrix} 1 & 3 & 9 \\ 3 & -2 & 1 \end{Bmatrix} \xrightarrow{l_2 - 3l_1} \begin{Bmatrix} 1 & 3 & 9 \\ 0 & -11 & -26 \end{Bmatrix}$$

$$\xrightarrow{l_2 \div -11} \begin{Bmatrix} 1 & 3 & 9 \\ 0 & 1 & 2 \end{Bmatrix} \xrightarrow{l_1 - 3l_2} \begin{Bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \end{Bmatrix}$$

$$\xrightarrow{l_1 - 3l_2} \begin{Bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \end{Bmatrix} \rightarrow \begin{cases} x = 3 \\ y = 2 \end{cases}$$

$$\begin{aligned} x - 3/2y &= 0 & \rightarrow & x - 3/2 \cdot 2 = 0 & \rightarrow & x = 3 \\ y &= 2 & & & & y = 2 \end{aligned}$$

$$b) \begin{cases} 5x + 8y = 34 \\ 10x + 16y = 50 \end{cases} \rightarrow \begin{Bmatrix} 5 & 8 & 34 \\ 10 & 16 & 50 \end{Bmatrix} \xrightarrow{l_2 - 2l_1} \begin{Bmatrix} 5 & 8 & 34 \\ 0 & 0 & -25 \end{Bmatrix}$$

$$\xrightarrow{l_1 \div 5} \begin{Bmatrix} 1 & 8/5 & 34/5 \\ 0 & 0 & -25 \end{Bmatrix} \rightarrow \begin{cases} x + 8/5y = 34/5 \\ 0 = -25 \end{cases}$$

Si $\rightarrow$ Sistema Impossível

$$c) \begin{cases} x + 2y = 5 \\ 2x - 3y = -4 \end{cases} \rightarrow \begin{Bmatrix} 1 & 2 & 5 \\ 2 & -3 & -4 \end{Bmatrix} \xrightarrow{l_2 - 2l_1} \begin{Bmatrix} 1 & 2 & 5 \\ 0 & -7 & -14 \end{Bmatrix}$$

$$\xrightarrow{l_2 \div -7} \begin{Bmatrix} 1 & 2 & 5 \\ 0 & 1 & 2 \end{Bmatrix} \xrightarrow{l_1 - 2l_2} \begin{Bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \end{Bmatrix}$$

$$\begin{cases} x = 1 \\ y = 2 \end{cases}$$

$$d) \begin{cases} 3x + 2y - 5z = 8 \\ 2x - 4y - 2z = -4 \\ x - 2y - 3z = -4 \end{cases} \rightarrow \begin{cases} 3 & 2 & -5 & 8 \\ 2 & -4 & -2 & -4 \\ 1 & -2 & -3 & -4 \end{cases} \xrightarrow{\text{troca } l_1 \text{ por } l_3} \begin{cases} 1 & -2 & -3 & -4 \\ 2 & -4 & -2 & -4 \\ 3 & 2 & -5 & 8 \end{cases}$$

$$\begin{cases} 1 & -2 & -3 & -4 \\ 0 & 0 & 4 & 4 \\ 0 & 8 & 4 & 20 \end{cases} \xrightarrow{l_2 \text{ por } l_3} \begin{cases} 1 & -2 & -3 & -4 \\ 0 & 8 & 4 & 20 \\ 0 & 0 & 4 & 4 \end{cases} \xrightarrow{l_1 + l_3} \begin{cases} 1 & -2 & 1 & 0 \\ 0 & 8 & 4 & 20 \\ 0 & 0 & 4 & 4 \end{cases} \xrightarrow{l_2 \div 4} \begin{cases} 1 & -2 & 1 & 0 \\ 0 & 8 & 4 & 20 \\ 0 & 0 & 1 & 1 \end{cases}$$

$$\begin{cases} 1 & -2 & 1 & 0 \\ 0 & 2 & 1 & 5 \\ 0 & 0 & 1 & 1 \end{cases} \xrightarrow{l_1 \leftarrow l_1 + l_2} \begin{cases} 1 & 0 & 2 & 5 \\ 0 & 2 & 1 & 5 \\ 0 & 0 & 1 & 1 \end{cases} \xrightarrow{l_2 \leftarrow l_2 + (-1)l_3} \begin{cases} 1 & 0 & 2 & 5 \\ 0 & 2 & 0 & 4 \\ 0 & 0 & 1 & 1 \end{cases}$$

$$\begin{cases} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{cases} \rightarrow \begin{matrix} x = 3 \\ y = 2 \\ z = 1 \end{matrix}$$

$$e) \begin{cases} 2x - 6y - 4z = -4 \\ x + 3y = 1 \\ 4x + 12y = 2 \end{cases} \rightarrow \begin{cases} 2 & -6 & -4 \\ 1 & 3 & 1 \\ 4 & 12 & 2 \end{cases} \xrightarrow{\text{troca } l_1 \text{ por } l_2} \begin{cases} 1 & 3 & 1 \\ 2 & -6 & -4 \\ 4 & 12 & 2 \end{cases}$$

$$\begin{cases} 1 & 3 & 1 \\ 0 & -12 & -6 \\ 0 & 0 & -2 \end{cases} \xrightarrow{l_2 \div -6} \begin{cases} 1 & 3 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & -2 \end{cases} \xrightarrow{l_2 \leftarrow l_2 + l_3} \begin{cases} 1 & 3 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & -2 \end{cases} \xrightarrow{l_3 \div -2} \begin{cases} 1 & 3 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{cases}$$

$$\begin{cases} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{cases} \rightarrow \text{Sistema Impossível}$$



$$f) \begin{cases} x + 2y - z = -2 \\ 2x - y + 3z = 9 \\ 3x + 3y - 2z = 3 \end{cases} \rightarrow \begin{cases} 1 & 2 & -1 & -2 \\ 2 & -1 & 3 & 9 \\ 3 & 3 & -2 & 3 \end{cases} \xrightarrow{\substack{l_2 + 2l_1 \\ l_3 + 3l_1}} \begin{cases} 1 & 2 & -1 & -2 \\ 0 & -5 & 5 & 5 \\ 0 & -3 & 1 & -3 \end{cases} \xrightarrow{-5} \begin{cases} 1 & 2 & -1 & -2 \\ 0 & 1 & -1 & -1 \\ 0 & -3 & 1 & -3 \end{cases}$$

$$\begin{cases} 1 & 2 & -1 & -2 \\ 0 & 1 & -1 & -1 \\ 0 & -3 & 1 & -3 \end{cases} \xrightarrow{\substack{l_1 + l_2 \\ l_3 + 3l_2}} \begin{cases} 1 & 0 & 0 & -4 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & -2 & -6 \end{cases} \xrightarrow{\div -2} \begin{cases} 1 & 0 & 0 & -4 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & 3 \end{cases} \xrightarrow{\substack{l_1 + 4l_3 \\ l_2 + l_3}} \begin{cases} 1 & 0 & 0 & 8 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 3 \end{cases}$$

$$\begin{cases} 1 & 0 & 0 & 8 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 3 \end{cases} \quad \begin{matrix} x = 8 \\ y = 4 \\ z = 3 \end{matrix}$$

$$g) \begin{cases} x + 0y + 3z = -8 \\ 2x - y + 0z = -4 \\ 3x - 2y - 5z = 26 \end{cases} \rightarrow \begin{cases} 1 & 0 & 3 & -8 \\ 2 & -1 & 0 & -4 \\ 3 & -2 & -5 & 26 \end{cases} \xrightarrow{\div 2} \begin{cases} 1 & 0 & 3 & -8 \\ 1 & -2 & 0 & -2 \\ 3 & -2 & -5 & 26 \end{cases} \xrightarrow{\substack{l_2 + l_1 \\ l_3 - 3l_1}} \begin{cases} 1 & 0 & 3 & -8 \\ 1 & -2 & 0 & -2 \\ 0 & -8 & -14 & 50 \end{cases}$$

$$\begin{cases} 1 & 0 & 3 & -8 \\ 0 & -2 & -3 & 16 \\ 0 & -8 & -14 & 50 \end{cases} \xrightarrow{\substack{l_2 + l_3 \\ l_3 + 3l_2}} \begin{cases} 1 & 0 & 3 & -8 \\ 0 & -2 & -3 & 16 \\ 0 & -2 & -11 & 50 \end{cases} \xrightarrow{\div -2} \begin{cases} 1 & 0 & 3 & -8 \\ 0 & 1 & 1.5 & -8 \\ 0 & 0 & -11 & 24 \end{cases}$$

$$\begin{cases} 1 & 0 & 3 & -8 \\ 0 & 1 & 1.5 & -8 \\ 0 & 0 & -11 & 24 \end{cases} \xrightarrow{\div -11} \begin{cases} 1 & 0 & 3 & -8 \\ 0 & 1 & 1.5 & -8 \\ 0 & 0 & 1 & -2.18 \end{cases} \xrightarrow{\substack{l_1 - 3l_3 \\ l_2 - 1.5l_3}} \begin{cases} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -2.18 \end{cases}$$

$$\begin{cases} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -2.18 \end{cases} \xrightarrow{\div -2.18} \begin{cases} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -2.18 \end{cases}$$

$$\begin{cases} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -2.18 \end{cases} \quad \begin{matrix} x = 4 \\ y = 3 \\ z = -2.18 \end{matrix}$$

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$$h) \begin{cases} x+2y+3z=10 \\ 3x+4y+6z=23 \\ 2x+2y+3z=13 \end{cases} \rightarrow \begin{cases} 1 & 2 & 3 & 10 \\ 3 & 4 & 6 & 23 \\ 2 & 2 & 3 & 13 \end{cases} \begin{cases} l_1 \div 1 \\ l_2 - 3l_1 \\ l_3 - 2l_1 \end{cases} \rightarrow \begin{cases} 1 & 2 & 3 & 10 \\ 0 & -2 & -3 & -7 \\ 0 & -2 & -3 & 13 \end{cases} \begin{cases} l_2 + l_1 \\ l_3 - l_1 \end{cases}$$

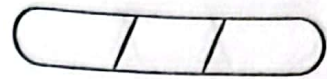
$$\begin{cases} -1 & -2 & -3 & -10 \\ 0 & -2 & -3 & -7 \\ 2 & 2 & 3 & 13 \end{cases} \begin{cases} l_1 + l_2 \\ l_3 + l_2 \end{cases} \rightarrow \begin{cases} -2 & -4 & -6 & -20 \\ 0 & -2 & -3 & -7 \\ 0 & -2 & -3 & -4 \end{cases} \begin{cases} l_1 + l_2 \div -2 \\ l_2 + l_1 \div -1 \\ l_3 + l_3 \div 1 \end{cases} \rightarrow \begin{cases} 1 & 2 & 3 & 10 \\ 0 & -2 & -3 & -7 \\ 0 & 0 & 0 & -7 \end{cases} \begin{cases} l_1 + l_2 \div -1 \\ l_3 + l_3 \div -1 \end{cases}$$

$$\begin{cases} 1 & 0 & 0 & 3 \\ 0 & 1 & 3 & 7/2 \\ 0 & 0 & 0 & -7 \end{cases} \begin{cases} x=3 \\ y=7/2-3/2z \\ z=1 \end{cases}$$

$$1) \begin{cases} x-3y+4z-w=2 \\ 2x-y+3z-2w=19 \end{cases} \rightarrow \begin{cases} 1 & -3 & 4 & -1 & 2 \\ 2 & -1 & 3 & -2 & 19 \end{cases} \begin{cases} l_2 + 2l_1 \\ l_2 - l_1 \end{cases} \rightarrow \begin{cases} -2 & 6 & -8 & 2 & -4 \\ 2 & -1 & 3 & -2 & 19 \end{cases} \begin{cases} l_2 + l_1 \end{cases}$$

$$\begin{cases} -2 & 6 & -8 & 2 & -4 \\ 0 & 5 & -5 & 0 & 15 \end{cases} \div 5 \rightarrow \begin{cases} -2 & 6 & -8 & 2 & -4 \\ 0 & 1 & -1 & 0 & 3 \end{cases} \begin{cases} l_1 + l_2 \\ x=6 \end{cases} \rightarrow \begin{cases} -2 & 0 & -2 & 2 & -2 \\ 0 & 1 & -1 & 0 & 3 \end{cases} \begin{cases} l_1 \div -2 \\ l_1 + l_2 \div -1 \end{cases}$$

$$\begin{cases} x_1 & x_2 & x_3 & x_4 \\ 1 & 0 & 1 & -1 & 11 \\ 0 & 1 & -1 & 0 & 3 \end{cases} \begin{cases} x=11+x_3-x_4 \\ y=3+x_3 \end{cases}$$



9) a) $\left(\begin{array}{ccc|c} 1 & 2 & 3 & 18 \\ 1 & 3 & 0 & 17 \\ 1 & 0 & 2 & 3 \end{array} \right) \xrightarrow{\substack{l_2 \leftarrow l_2 - l_1 \\ l_3 \leftarrow l_3 - l_1}} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 18 \\ 0 & 1 & -3 & 0 \\ 0 & -2 & -1 & -5 \end{array} \right) \xrightarrow{l_3 \leftarrow l_3 + 2l_2} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 18 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & -7 & -5 \end{array} \right)$

$\xrightarrow{l_1 \leftarrow l_1 - 2l_2} \left(\begin{array}{ccc|c} 1 & 0 & 9 & 10 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & -7 & -5 \end{array} \right) \xrightarrow{l_1 \leftarrow l_1 + 9l_3} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & -7 & -5 \end{array} \right) \xrightarrow{l_2 \leftarrow l_2 + 3l_3} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & -7 & -5 \end{array} \right) \xrightarrow{\div -7} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{array} \right)$

$a = 1 - d \quad // \quad b = 2 \quad // \quad c = 1$

b) $\left(\begin{array}{ccc|c} 1 & 1 & 3 & -3 \\ 0 & 2 & 1 & -3 \\ 1 & 0 & 2 & -1 \end{array} \right) \xrightarrow{l_3 \leftarrow l_3 - l_1} \left(\begin{array}{ccc|c} 1 & 1 & 3 & -3 \\ 0 & 2 & 1 & -3 \\ 0 & -1 & -1 & 2 \end{array} \right) \xrightarrow{l_2 \leftrightarrow l_3} \left(\begin{array}{ccc|c} 1 & 1 & 3 & -3 \\ 0 & -1 & -1 & 2 \\ 0 & 2 & 1 & -3 \end{array} \right) \xrightarrow{l_2 \times -1} \left(\begin{array}{ccc|c} 1 & 1 & 3 & -3 \\ 0 & 1 & 1 & -2 \\ 0 & 2 & 1 & -3 \end{array} \right)$

$\xrightarrow{l_1 \leftarrow l_1 - l_2} \left(\begin{array}{ccc|c} 1 & 0 & 2 & -1 \\ 0 & 1 & 1 & -2 \\ 0 & 2 & 1 & -3 \end{array} \right) \xrightarrow{l_3 \leftarrow l_3 - 2l_2} \left(\begin{array}{ccc|c} 1 & 0 & 2 & -1 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & -1 & 1 \end{array} \right) \xrightarrow{l_1 \leftarrow l_1 + 2l_3} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & -1 & 1 \end{array} \right) \xrightarrow{l_2 \leftarrow l_2 + l_3} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & -1 & 1 \end{array} \right)$

$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 \end{array} \right) \quad \begin{array}{l} a = 1 - d \\ b = 2 + d \\ c = -1 + d \end{array}$

3)

$$5) a) (A+4I_3)X = 0 \quad \begin{pmatrix} 1 & 0 & 5 \\ 1 & 1 & 1 \\ 0 & 1 & -1 \end{pmatrix} + 4 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 5 \\ 1 & 1 & 1 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 5 & 0 & 5 \\ 1 & 5 & 1 \\ 0 & 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} 5 & 0 & 5 & 0 \\ 1 & 5 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \div 5 \rightarrow \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 5 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \begin{matrix} \\ l_2 \leftarrow l_2 - l_1 \\ \end{matrix}$$

$$\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \div 5 \rightarrow \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{matrix} \\ \\ l_3 \leftarrow l_3 + (-1)l_2 \end{matrix}$$

$$\begin{matrix} X = -Z // Y = 0 \\ X = -a // Y = 0 // Z = a \end{matrix} \quad \begin{cases} -a \\ 0 \\ a \end{cases}$$

$$b) AX = 2X \rightarrow AX - 2X = 0 \rightarrow (A - 2I)X = 0$$

$$\begin{pmatrix} 1 & 0 & 5 \\ 1 & 1 & 1 \\ 0 & 1 & -1 \end{pmatrix} - 2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot X = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 5 \\ 1 & 1 & 1 \\ 0 & 1 & -1 \end{pmatrix} + \begin{pmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} -1 & 0 & 5 \\ 1 & -1 & 1 \\ 0 & 1 & -6 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 0 & 5 & 0 \\ 1 & -1 & 1 & 0 \\ 0 & 1 & -6 & 0 \end{pmatrix} \begin{matrix} \\ l_2 \leftarrow l_2 + l_3 \\ \\ \end{matrix} \rightarrow \begin{pmatrix} -1 & 0 & 5 & 0 \\ 0 & -6 & 0 \\ 0 & 1 & -6 & 0 \end{pmatrix} \begin{matrix} \\ \\ l_3 \leftarrow l_3 + l_2 \end{matrix} \rightarrow \begin{pmatrix} -1 & 0 & 5 & 0 \\ 0 & -6 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{matrix} \\ \div -1 \\ \div -1 \end{matrix} \rightarrow \begin{pmatrix} 1 & 0 & -5 & 0 \\ 0 & 1 & -6 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\begin{cases} 1 & 0 & -5 & 0 \\ 0 & 1 & -6 & 0 \\ 0 & 0 & 0 & 0 \end{cases} \rightarrow \begin{cases} X - 5Z = 0 \\ Y - 6Z = 0 \\ 0 = 0 \end{cases} \rightarrow \begin{cases} X = 5Z \\ Y = 6Z \\ Z = a \end{cases} \rightarrow \begin{cases} X = 5a \\ Y = 6a \\ Z = a \end{cases}$$

6) a) $\begin{cases} X + Y + Z = 2 \\ 2X + 3Y + 2Z = 5 \\ 2X + 3Y + (a^2 - 1)Z = a + 1 \end{cases} \rightarrow \begin{cases} 1 & 1 & 1 & 2 \\ 2 & 3 & 2 & 5 \\ 2 & 3 & a^2 - 1 & a + 1 \end{cases}$

$\begin{cases} l_2 + l_1 + (-2l_1) \\ l_3 + l_1 + (-2l_1) \end{cases}$

$$\begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & a^2 - 3 & a - 1 \end{pmatrix} \xrightarrow{\begin{matrix} l_1 + l_2 + (-l_2) \\ l_3 + l_2 + (-l_2) \end{matrix}} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & a^2 - 3 & a - 1 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}$$

$\text{SPD} = \text{POSTO de } A \neq \text{POSTO de } AP = N_{\text{incompat}} = 3$

$a^2 - 3 \neq 0$ e $a - 1 \neq 0$ $a \neq \pm\sqrt{3}$ e $a \neq 1$

$\text{SP1} = \text{POSTO de } A = \text{POSTO de } AP < N_{\text{incompat}} = 3$

2 $a^2 - 3 = 0$ 2 SPD 3

Não existe um número que satisfaça a

SI $\text{POSTO de } A \neq \text{POSTO de } AP$

3 e 2 ou 2 e 3

$A = \pm\sqrt{3}$ e $a = 1$

$$b) \begin{cases} x + 2y - 3z = 4 \\ 3x - y + 5z = 2 \\ 4x + y + (a^2 - 14)z = a + 2 \end{cases} \rightarrow \begin{cases} 1 & 2 & -3 & 4 \\ 3 & -1 & 5 & 2 \\ 4 & 1 & a^2 - 14 & a + 2 \end{cases} \begin{cases} l_2 \leftarrow l_2 + (-3l_1) \\ l_3 \leftarrow l_3 + (-4l_1) \end{cases}$$

$$\begin{cases} 1 & 2 & -3 & 4 \\ 0 & -7 & 14 & -10 \\ 0 & -7 & a^2 - 2 & a - 14 \end{cases} \xrightarrow{l_3 \leftarrow l_3 + (-l_2)} \begin{cases} 1 & 2 & -3 & 4 \\ 0 & -7 & 14 & -10 \\ 0 & 0 & a^2 - 16 & a + 2 \end{cases} \xrightarrow{l_2 \leftarrow l_2 \cdot (-1/7)} \begin{cases} 1 & 2 & -3 & 4 \\ 0 & 1 & -2 & 10/7 \\ 0 & 0 & a^2 - 16 & a + 2 \end{cases}$$

IA e AP

$$\begin{pmatrix} 7 & 0 & -7 & 8 \\ 0 & -14 & 28 & -20 \\ 0 & 0 & a^2 - 16 & a + 2 \end{pmatrix} \xrightarrow{\substack{\div 7 \\ \div -14}} \begin{pmatrix} 1 & 0 & -1 & 8/7 \\ 0 & 1 & -2 & 10/7 \\ 0 & 0 & a^2 - 16 & a + 2 \end{pmatrix}$$

$$SPD = \underset{3}{\text{POSTO } A} = \underset{3}{\text{POSTO } AP} = \underset{3}{N \text{ de Independentes}}$$

$$\text{Proviso } a^2 - 16 \neq 0 \parallel a - 2 \neq 0 \rightarrow a \neq 4 \text{ e } -2$$

$$SPI = \underset{2}{\text{POSTO } A} - \underset{2}{\text{POSTO } AP} < \underset{3}{N \text{ de Independentes}}$$

$$a^2 - 16 = 0 \text{ e } a - 2 = 0 \parallel \text{Logo } a = 2$$

$$SI = \underset{3}{\text{POSTO } A} \neq \underset{2 \text{ e } 3}{\text{POSTO de } AP}$$

$$a^2 - 16 = 0 \text{ e } a - 2 \neq 0 \parallel \text{ou } a^2 - 16 \neq 0 \text{ e } a - 2 = 0 \\ a = -2 \text{ ter\'a SI}$$

$$M = [A | I_n] \rightarrow \tilde{M} = [I_n | A^{-1}]$$

$$7a) \begin{pmatrix} 2 & -2 \\ 3 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & -2 & 1 & 0 \\ 3 & 1 & 0 & 1 \end{pmatrix} \begin{matrix} l_1 + l_1 \cdot 3 \\ l_2 + l_2 \cdot 2 \end{matrix} \rightarrow \begin{pmatrix} 6 & -6 & 3 & 0 \\ 6 & -2 & 0 & 2 \end{pmatrix} \begin{matrix} \\ l_2 + l_2 + (-l_1) \end{matrix}$$

$$\begin{pmatrix} 6 & -6 & 3 & 0 \\ 0 & 8 & -3 & 2 \end{pmatrix} \begin{matrix} \times 4 \\ \times 3 \end{matrix} \rightarrow \begin{pmatrix} 24 & -24 & 12 & 0 \\ 0 & 24 & -9 & 6 \end{pmatrix} \begin{matrix} l_1 + l_1 \cdot l_2 \\ \\ \end{matrix} \rightarrow \begin{pmatrix} 24 & 0 & 3 & 6 \\ 0 & 24 & -9 & 6 \end{pmatrix} \begin{matrix} \div 24 \\ \div 24 \end{matrix}$$

$$\begin{pmatrix} 1 & 0 & 1/8 & 1/4 \\ 0 & 1 & -3/8 & 1/4 \end{pmatrix} \quad A^{-1} = \begin{pmatrix} 1/8 & 1/4 \\ -3/8 & 1/4 \end{pmatrix}$$

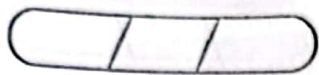
$$b) \begin{pmatrix} 2 & -2 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & -2 & 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 & 0 & 1 \end{pmatrix} \begin{matrix} l_2 \leftrightarrow l_1 \\ l_1 \leftrightarrow l_2 \\ \end{matrix} \rightarrow \begin{pmatrix} 1 & 2 & 1 & 0 & 1 & 0 \\ 2 & -2 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 1 \end{pmatrix} \begin{matrix} \\ l_2 + l_2 + (-l_1) \\ \end{matrix}$$

$$\begin{pmatrix} 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & -6 & -2 & 1 & -2 & 0 \\ 0 & 1 & -1 & 0 & 0 & 1 \end{pmatrix} \begin{matrix} l_1 + l_1 \cdot 3 \\ \\ l_3 + l_3 \cdot 6 \end{matrix} \rightarrow \begin{pmatrix} 3 & 6 & 3 & 0 & 3 & 0 \\ 0 & -6 & -2 & 1 & -2 & 0 \\ 0 & 6 & -6 & 0 & 6 & 6 \end{pmatrix} \begin{matrix} l_1 + l_1 + l_2 \\ \\ l_3 + l_3 + l_2 \end{matrix}$$

$$\begin{pmatrix} 3 & 0 & 1 & 1 & 1 & 0 \\ 0 & -6 & -2 & 1 & -2 & 0 \\ 0 & 0 & -8 & 1 & -2 & 6 \end{pmatrix} \begin{matrix} l_1 + l_1 \cdot 8 \\ l_2 + l_2 \cdot 3 \\ \end{matrix} \rightarrow \begin{pmatrix} 24 & 0 & 8 & 8 & 8 & 0 \\ 0 & -24 & -8 & 4 & -8 & 0 \\ 0 & 0 & -8 & 1 & -2 & 6 \end{pmatrix} \begin{matrix} l_1 + l_1 + l_3 \\ l_2 + l_2 + (-l_3) \\ \end{matrix}$$

$$\begin{pmatrix} 24 & 0 & 0 & 9 & 6 & 6 \\ 0 & -24 & 0 & 3 & -6 & -6 \\ 0 & 0 & -8 & 1 & -2 & 6 \end{pmatrix} \begin{matrix} \div 24 \\ \div -24 \\ \div -8 \end{matrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 & 3/8 & 1/4 & 1/4 \\ 0 & 1 & 0 & -1/8 & 1/4 & 1/4 \\ 0 & 0 & 1 & -1/8 & 1/4 & -3/4 \end{pmatrix}$$

$$A^{-1} = \begin{pmatrix} 3/8 & 1/4 & 1/4 \\ -1/8 & 1/4 & 1/4 \\ -1/8 & 1/4 & -3/4 \end{pmatrix}$$



$$c) \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix} \rightarrow \begin{pmatrix} 3 & 5 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{pmatrix} \xrightarrow{l_1 \leftrightarrow l_2} \begin{pmatrix} 1 & 2 & 0 & 1 \\ 3 & 5 & 1 & 0 \end{pmatrix} \xrightarrow{l_2 \leftarrow l_2 + (-3)l_1}$$

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & -1 & -3 \end{pmatrix} \xrightarrow{l_1 \leftarrow l_1 + (2)l_2} \begin{pmatrix} 1 & 0 & -2 & 5 \\ 0 & -1 & -3 \end{pmatrix} \xrightarrow{\times -1} \begin{pmatrix} 1 & 0 & -2 & 5 \\ 0 & 1 & 3 \end{pmatrix}$$

$$A^{-1} = \begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix}$$

$$d) \begin{pmatrix} 0 & -1 & 1 \\ 2 & 0 & -1 \\ 1 & 1 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & -1 & 1 & 1 & 0 & 0 \\ 2 & 0 & -1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{\substack{l_1 \leftrightarrow l_3 \\ l_2 \leftrightarrow l_2 + (-2)l_3}} \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 1 \\ 2 & 0 & -1 & 0 & 1 & 0 \\ 0 & -1 & 1 & 1 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & -2 & -1 & 0 & 1 & -2 \\ 0 & -1 & 1 & 1 & 0 & 0 \end{pmatrix} \xrightarrow{\substack{l_1 \leftarrow l_1 \times 2 \\ l_3 \leftarrow l_3 \times -2}} \begin{pmatrix} 2 & 2 & 0 & 0 & 0 & 2 \\ 0 & -2 & -1 & 0 & 1 & -2 \\ 0 & 2 & -2 & -2 & 0 & 0 \end{pmatrix} \xrightarrow{\substack{l_1 \leftarrow l_1 + l_2 \\ l_3 \leftarrow l_3 + l_2}}$$

$$\begin{pmatrix} 2 & 0 & -1 & 0 & 1 & 0 \\ 0 & -2 & -1 & 0 & 1 & -2 \\ 0 & 0 & -3 & -2 & 1 & -2 \end{pmatrix} \xrightarrow{\substack{\times -3 \\ \times -3}} \begin{pmatrix} -6 & 0 & 3 & 0 & -3 & 0 \\ 0 & 6 & 3 & 0 & -3 & 6 \\ 0 & 0 & -3 & -2 & 1 & -2 \end{pmatrix} \xrightarrow{\substack{l_1 \leftrightarrow l_3 \\ l_2 \leftrightarrow l_2}} \begin{pmatrix} 0 & 6 & 3 & 0 & -3 & 6 \\ -6 & 0 & 0 & -2 & -2 & -2 \\ 0 & 0 & -3 & -2 & 1 & -2 \end{pmatrix} \xrightarrow{\substack{\div -6 \\ \div 6 \\ \div -3}}$$

$$\begin{pmatrix} 1 & 0 & 0 & 1/3 & -1/3 & 1/3 \\ 0 & 1 & 0 & -1/3 & -1/3 & 2/3 \\ 0 & 0 & 1 & 2/3 & -1/3 & 2/3 \end{pmatrix}$$

$$A^{-1} = \begin{pmatrix} 1/3 & 1/3 & 1/3 \\ -1/3 & -1/3 & 2/3 \\ 2/3 & -1/3 & 2/3 \end{pmatrix}$$

$$e) \begin{pmatrix} 2 & -1 & 0 & 1 \\ 0 & 2 & 0 & -1 \\ 1 & 0 & 2 & 3 \\ 0 & 0 & -2 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & -1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & -1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 2 & 3 & 0 & 0 & 1 & 0 \\ 0 & 0 & -2 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{matrix} l_1 \\ l_2 \\ l_3 \end{matrix} \rightarrow \begin{pmatrix} 1 & 0 & 2 & 3 & 0 & 0 & 1 & 0 \\ 0 & 2 & 0 & -1 & 0 & 1 & 0 & 0 \\ 2 & -1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & -2 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 2 & 3 & 0 & 0 & 1 & 0 \\ 0 & 2 & 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & 0 & 0 & 0 & 0 & 1 \\ 2 & -1 & 0 & 1 & 1 & 0 & 0 & 0 \end{pmatrix} \begin{matrix} l_1 + l_3 - 2l_2 \\ l_4 - 2l_1 \end{matrix} \rightarrow \begin{pmatrix} 1 & 0 & 2 & 3 & 0 & 0 & 1 & 0 \\ 0 & 2 & 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & 0 & 0 & 0 & 0 & 1 \\ 0 & -1 & -4 & 5 & 1 & 0 & -2 & 0 \end{pmatrix} \begin{matrix} l_4 \times 2 \end{matrix}$$

$$\begin{pmatrix} 1 & 0 & 2 & 3 & 0 & 0 & 1 & 0 \\ 0 & 2 & 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & -8 & 11 & 2 & 1 & -4 & 0 \end{pmatrix} \begin{matrix} l_1 + l_2 \\ l_4 + l_3 \end{matrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 & 3 & 0 & 0 & 1 & 1 \\ 0 & 2 & 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 8 & 0 & 0 & 0 & 0 & -4 \\ 0 & 0 & -8 & 11 & 2 & 1 & -4 & 0 \end{pmatrix} \begin{matrix} l_1 \times 2 \\ l_4 + l_3 \end{matrix}$$

$$\begin{pmatrix} -11 & 0 & 0 & -33 & 0 & 0 & -11 & -11 \\ 0 & 22 & 0 & -11 & 0 & 11 & 0 & 0 \\ 0 & 8 & 0 & 0 & 0 & 0 & 0 & -4 \\ 0 & 0 & 0 & 11 & -2 & -1 & 4 & 4 \end{pmatrix} \begin{matrix} l_1 + l_2 + (3l_3) \\ l_2 + l_4 \end{matrix} \rightarrow \begin{pmatrix} -11 & 0 & 0 & 0 & 0 & 0 & 11 & 11 \\ 0 & 22 & 0 & 0 & 0 & 0 & -2 & 10 \\ 0 & 8 & 0 & 0 & 0 & 0 & 0 & -4 \\ 0 & 0 & 0 & 11 & -2 & -1 & 4 & 4 \end{pmatrix} \begin{matrix} \div -11 \\ \div 22 \\ \div 8 \\ \div 11 \end{matrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{matrix} +6/11 & 3/11 & -1/11 & -1/11 \\ -1/11 & 5/11 & 2/11 & 3/11 \\ 0 & 0 & 0 & -1/2 \\ -2/11 & -1/11 & 2/11 & 2/11 \end{matrix}$$

8) $1c + 2s + 3b = 26$
 $2c + 5s + 6b = 60 \rightarrow$
 $2c + 3s + 4b = 40$

$$\begin{pmatrix} 1 & 2 & 3 & 26 \\ 2 & 5 & 6 & 60 \\ 2 & 3 & 4 & 40 \end{pmatrix} \begin{array}{l} \\ l_2 + l_2 + (-2l_1) \\ l_3 + l_3 + (-2l_1) \end{array}$$

$$\begin{pmatrix} 1 & 2 & 3 & 26 \\ 0 & 1 & 0 & 8 \\ 0 & -1 & -2 & -12 \end{pmatrix} \begin{array}{l} l_1 + l_1 + (-2l_2) \\ \\ l_3 + l_3 + l_2 \end{array} \rightarrow \begin{pmatrix} 1 & 0 & 3 & 10 \\ 0 & 1 & 0 & 8 \\ 0 & 0 & -2 & -4 \end{pmatrix} \begin{array}{l} \\ \\ l_3 + l_3 \div -2 \end{array}$$

$$\begin{pmatrix} 1 & 0 & 3 & 10 \\ 0 & 1 & 0 & 8 \\ 0 & 0 & 1 & 2 \end{pmatrix} \begin{array}{l} l_1 + l_1 + (3l_3) \\ \\ \end{array} \rightarrow \begin{pmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 8 \\ 0 & 0 & 1 & 2 \end{pmatrix} \begin{array}{l} Calça = 4 \\ Short = 8 \\ blusa = 2 \end{array}$$

9) $\begin{cases} X = 3Y \\ X = Y + Z \\ 3Y = Y + Z \end{cases} \rightarrow \begin{cases} 2X + 6Y + 5Z = 2200 \end{cases}$ Substituindo

$2Y = Z$

$\begin{cases} (2 \cdot 3Y) + 6Y + (5 \cdot 2Y) = 2200 \\ 22Y = 2200 \end{cases} \rightarrow Y = 100$

$X = 3Y \rightarrow X = 3 \cdot 100 \rightarrow X = 300$
 $2Y = Z \rightarrow 2 \cdot 100 = Z \rightarrow Z = 200$
 $Y = 100$

$(2 \cdot 300) + (6 \cdot 100) + (5 \cdot 200) = 2200$
 $2200 = 2200$

10) $10C + 30S + 10P = 7000$
 $20C + 40S + 20P = 6000 \rightarrow$
 $10C + 20S + 10P = 5000$

$$\begin{pmatrix} 1 & 2 & 3 & 700 \\ 2 & 4 & 2 & 600 \\ 1 & 2 & 1 & 500 \end{pmatrix} \begin{array}{l} \\ l_2 + l_2 + (-2l_1) \\ l_3 + l_3 + (-l_1) \end{array}$$

$$\begin{pmatrix} 1 & 2 & 3 & 700 \\ 0 & 0 & -2 & -100 \\ 0 & 0 & -2 & -200 \end{pmatrix} \begin{array}{l} \\ \\ l_3 + l_3 + l_2 \end{array}$$

$$\begin{pmatrix} 1 & 2 & 3 & 700 \\ 0 & 0 & -2 & -100 \\ 0 & 0 & 0 & -300 \end{pmatrix}$$

$S = 100$
 $P = 80$
 $C = 140$

tilibra

$$(11) \begin{pmatrix} 2 & 3 & 1 & 8420 \\ 1 & 2 & 2 & 7940 \\ 4 & 3 & 0 & 8110 \end{pmatrix} \xrightarrow{l_1 \leftrightarrow l_2} \begin{pmatrix} 1 & 2 & 2 & 7940 \\ 2 & 3 & 1 & 8420 \\ 4 & 3 & 0 & 8110 \end{pmatrix} \xrightarrow{\begin{matrix} l_2 \leftarrow l_2 - 2l_1 \\ l_3 \leftarrow l_3 - 4l_1 \end{matrix}} \begin{pmatrix} 1 & 2 & 2 & 7940 \\ 0 & -1 & -3 & -7460 \\ 0 & -5 & -8 & -23650 \end{pmatrix} \xrightarrow{l_1 + l_2(2)} \begin{pmatrix} 1 & 0 & -4 & -6980 \\ 0 & -1 & -3 & -7460 \\ 0 & -5 & -8 & -23650 \end{pmatrix}$$

$$\xrightarrow{l_3 + l_3(-5l_2)} \begin{pmatrix} 1 & 0 & -4 & -6980 \\ 0 & -1 & -3 & -7460 \\ 0 & 0 & 7 & 13650 \end{pmatrix} \xrightarrow{\times} \begin{pmatrix} 1 & 0 & -4 & -6980 \\ 0 & 1 & 3 & 7460 \\ 0 & 0 & 7 & 13650 \end{pmatrix} \quad 7C = 13650 \quad C = 1950$$

$$B + 3C = 7460 \rightarrow B + 5850 = 7460 \rightarrow B = 1610$$

$$A - 7800 = -6980 \rightarrow A = 820$$

$$A = 820 \quad // \quad B = 1610 \quad // \quad C = 1950$$